PYTHON LEARNING: DAY 03

Statistics Essentials: Mean, Median & Mode (Logic + Library)

1. Sorting: The Basic Foundation

Median nikalne ke liye data ka order mein hona zaroori hai.

```
data = [12, 10, 15, 20, 11]

# Step 1: Data ko Sort karna (Zaroori Step)
data.sort()
# for descending order: data.sort(reverse=True)
print(f"Sorted Data: {data}")
```

Output: [10, 11, 12, 15, 20]

2. Median: Logic (Odd & Even Cases)

Case A: Odd Data (5 items)

```
data = [12, 10, 15, 20, 11]
data.sort()
n = len(data)
middle_index = n // 2

median = data[middle_index]
print(f"The Median is: {median}")
```

Output: 12

Case B: Even Data (4 items)

```
data = [12, 10, 15, 20]
data.sort() # [10, 12, 15, 20]
n = len(data)
mid1 = n // 2       # Index 2 (15)
mid2 = (n // 2) - 1 # Index 1 (12)

median = (data[mid1] + data[mid2]) / 2
print(f"The Median is: {median}")
```

Output: 13.5

3. Mode: Origin Logic (Dictionary & Max)

```
data = [18, 20, 18, 22, 18, 20, 24]

# Step 1: Har number ki ginti (Count) nikalna
ginti = {}
for i in data:
    ginti[i] = data.count(i)

# Step 2: Sabse badi ginti wala number dhoondhna
# 'key=ginti.get' values (counts) ke basis par check karta hai
mode_value = max(ginti, key=ginti.get)

print(f"Logic se nikala gaya Mode: {mode_value}")
```

Output: 18

4. Statistics Library: Professional Way

```
import statistics
data = [10, 20, 30, 40]

# Standard Median & Mode
print("Normal Median:", statistics.median(data)) # Output: 25.0

# Advanced Median (For Even Data)
print("Median Low:", statistics.median_low(data))   # Output: 20
print("Median High:", statistics.median_high(data)) # Output: 30

# Multi-Mode: Agar multiple items same bar repeat ho
# statistics.multimode(data) ka use karein
```

Pro Tip: `statistics.median()` itna smart hai ki wo Odd/Even apne aap handle kar leta hai aur data sort karne ki bhi tension nahi hoti.